



Integration of Hybrid GA-PSO-SA with Self-adaptive Dynamic Attention for Multi-objective CNN-based Oil Palm Fruit Ripeness Classification

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Abstract: Determining exactly when oil palm fruit is ripe is crucial for maximizing harvest efficiency, but current methods achieve only 65–75% accuracy, resulting in 15–20% annual losses. This research introduces a hybrid GA-PSO-SA metaheuristic optimization framework featuring self-adaptive dynamic attention for multi-objective CNN-based classification. The best CNN+GA+PSO setup used 10,000 photos from five Indonesian plantation sites, with four ripeness classifications (Unripe, Underripe, Ripe, Overripe). It achieved 95.73% accuracy and 95.84% F1-score, indicating a better balance between bias and variance with only a 0.16% precision-recall gap. The suggested technique achieves a mean test accuracy of $96.12\% \pm 0.39\%$ (max 96.86%) across 10 independent runs, with a narrow 95% confidence range [95.85%, 96.39%]. This is much better than baseline methods like MobileNetV2 (95.53%), ResNet50 (95.27%), and ViT-tiny (96.06%). The ablation study shows that dual-metaheuristic optimization is 18.23% better than single methods. However, adding more architectural complexity (SA, CBAM, SADA) worsens performance due to the curse of dimensionality. Leave-Site-Out cross-validation achieved $94.66\% \pm 0.54\%$ accuracy across previously unobserved locations, thereby validating spatial generalization. The Pareto-optimal solution has a latency of 2.61 ms and a model size of 43.6 MB, which is 4.1 times quicker than ViT-tiny. This enables the deployment of precision agriculture apps on edge devices in real time.

Keywords: Hybrid GA-PSO-SA, Multi-objective CNN, Oil palm fruit, Dynamic attention, Pareto-optimal.

1. Introduction

Figuring out the right time to harvest is a major challenge in the industry. Mistakes in judging fruit ripeness cause losses of 15-20 percent, costing billions of rupiah each year [1]. The current method depends on harvesters visually checking the fruit, which is only 65-75% accurate and varies a lot depending on each worker's experience. As a result, the quality is often inconsistent [2].

CNNs have been successful in agricultural image classification and reported satisfactory accuracies of 85–92% for multiple fruit commodities [3]. Nevertheless, the use of CNNs on oil palm fruits involves complex issues that are not well addressed. The VGG-16 model achieved an accuracy of 83.4%;

however, under strong lighting conditions and occlusion, it was unable to generalize properly. ResNet50 outperformed with 87.1% accuracy, but its inference time (2.3 seconds) is not fast enough for real-time field applications [4]. The MobileNetV2 approach is more efficient (0.4 seconds), but its accuracy drops dramatically to 79.2% on datasets with high variation [5].

Three fundamental gaps were identified from the literature review: First, conventional CNNs have difficulty extracting discriminative features in conditions of uneven lighting, extreme viewing angles, and partial occlusion by leaves or fronds-conditions that are very common in plantations. Existing models tend to overfit on training datasets and fail to generalize to field conditions [6]. Second, no research has systematically optimized CNN

hyperparameters for the specific palm oil domain. Most studies use default hyperparameters or inefficient and suboptimal manual trial-and-error tuning [7]. Third, the balance among accuracy, inference speed, and memory consumption has not been optimally addressed jointly [8].

Accurate models are slow and resource-hungry, but if efficiency comes at the expense of accuracy, it is unacceptable in field applications. Attention mechanisms have been found to enable CNNs to concentrate on informative regions, which achieves a 3-7% increase in fruit classification accuracy [3]. However, the static attention mechanism, which uses a single fixed weight for each input, is not very flexible in responding to changes in inputs, including those from oil palm fruits. It is possible that dynamic attention, which can fine-tune the focus strategy for each sample, could address these challenges; however, it has not been well studied in this setting. Optimal hyperparameter selection through metaheuristic algorithms, such as Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA), has been proven to enhance the performance of CNNs in numerous applications [9].

GA excels in global exploration but converges slowly; PSO is fast but easily gets stuck in local optima; SA is effective at escaping local optima but is sensitive to control parameters. Hybrid algorithms that adaptively combine the strengths of GA, PSO, and SA can produce more optimal solutions than single methods; however, they have not been applied to CNN optimization for oil palm fruit ripeness [10].

This research develops an integrated framework that addresses this gap through three main contributions: (1) A CNN architecture with a self-adaptive dynamic attention mechanism that automatically adjusts the attention strategy based on the complexity characteristics of each input image-providing more intensive attention to difficult samples and efficient attention to easy samples [11]; (2) A hybrid GA-PSO-SA algorithm for simultaneous optimization of multiple CNN hyperparameters (learning rate, batch size, dropout rate, number of layers, filter size) with an adaptive switching mechanism that leverages the strengths of each algorithm in different optimization phases [12]; (3) A true multi-objective optimization framework that explicitly optimizes three conflicting objectives: maximizing classification accuracy, minimizing inference latency, and minimizing model size through a Pareto-optimal solution set, resulting in an ensemble of solutions with different trade-offs that can be selected according to deployment needs (high

accuracy for quality control vs. high efficiency for edge devices) [9].

The integration of hybrid GA-PSO-SA with self-adaptive dynamic attention in this multi-objective optimization formulation is expected to produce solutions that are not only accurate but also practical for real-time implementation in the field, supporting increased operational efficiency of oil palm plantations and optimization of national palm oil production quality [13].

Paper Outline: The structure of the paper is as follows: First, we briefly review related works (Section 2), including Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Simulated Annealing (SA), and dynamic attention mechanisms, then some preprocessing methods and optimization algorithms for oil palm fruit quality classification. The research method, the combination model of Hybrid GA-PSO-SA and Self-Adaptive Dynamic Attention on multi-objective CNN-based oil palm fruit quality classification, data preprocessing (normalization, augmentation, and batch-size processing), and the feature fusion process for improving the accuracy of oil palm fruit recognition are offered in Section 3. Section 4 demonstrates the experimental results, evaluation metrics, and the merits and demerits of the proposed method. Summarizing and concluding remarks are given in Section 5 with the recommendations for future work.

2. Related work

The last decade has witnessed rapid developments in automatic classification using deep learning; however, numerous challenges remain in simultaneously optimizing multiple objectives and incorporating adaptive attention mechanisms. This chapter reviews the literature, highlights the limitations of traditional approaches, and lays the groundwork for this research.

Some studies have investigated the application of single metaheuristic algorithms to CNN optimization. The GA-based method has achieved better results in multiple domains, such as building heating load estimation and pattern classification [14]. However, GA has some inherent drawbacks in terms of speed of convergence and its propensity to get trapped by local optima in complex, multi-modal optimization landscapes. Moreover, GA needs large populations with sufficient iterations to obtain good quality solutions, which leads to an excessive amount of computational time [14].

As an optimization algorithm, PSO has been successfully applied to neural network hyperparameter optimization and has demonstrated

effectiveness in several application fields [15]. Nevertheless, PSO converges early and explores the search space not intensively enough, particularly for high-dimensional problems. Research has demonstrated that the only implementation of PSO generally performs worse than GA for certain cases [16].

Simulated Annealing (SA) provides a mechanism to escape local optima by probabilistically accepting inferior solutions. However, SA has vulnerable temperature schedule parameters that need to be carefully configured to achieve the best effectiveness. The shortcomings of SA are a slow convergence rate and a limited ability to gather global optimal solutions [17].

To address these limitations, combining a few metaheuristics has been considered in the literature. compared CNN-SA and CNN-PSO to a conventional CNN and demonstrated that the hybrid approach achieved better results. However, the study incorporated only two algorithms and did not consider the combination of three algorithms simultaneously [18]. For example, [14] has demonstrated that GA and PSO with artificial bee colony (GA-PSO-ABC) achieved superior results, whereas the optimization process was not focused on simultaneously optimizing several objectives.

Introduced PSO-CNN-SA, a model that combines a self-attention mechanism with hybrid optimization. The experimental results showed that their proposed model had better positioning accuracy in data fusion. Unfortunately, it did not address self-adaptive dynamic attention under a multi-objective optimization formulation [19]. A notable difference is that Wang and Li employed a static attention mechanism, whereas we incorporate a self-adaptive dynamic attention mechanism, which can vary the attending position according to different input information.

While attention mechanisms have been shown to be effective when embedded on top of a CNN for image and text classification, [20] found that a CNN with attention (CNN-ATN) offers better performance than a traditional CNN in SER. Nevertheless, the paper employed static attention without adapting to different inputs[20].

introduced the AC-CNN (Attention Causal Convolutional Neural Network), which features dynamically adjustable attention and residual connections, thereby accelerating the convergence process. Nonetheless, dynamic attention was not integrated with the hybrid metaheuristic optimizer, and the resulting problem was not transformed into a multi-objective optimization one [21]. The main difference from this work is that AC-CNN does not

utilize the GA-PSO-SA hybrid optimization method for concurrent hyperparameter tuning.

Integrating multi-objective optimization with classification remains one of the under-researched topics. [22] investigated multi-task learning for lung nodule categorization by employing the weighted fusion of multiple models. However, this approach did not explicitly optimize multiple objectives within a single model, nor did it utilize metaheuristic optimization [22].

Presented a three-objective-partner-banking sector selection system with GA and a multilayer perceptron. However, the research didn't hybridize multiple metaheuristic algorithms and did not apply them to image classification [23].

There is a lack of existing research on fruit ripeness classification using deep learning. [24] presented a stacking fusion model that focused on strawberry quality grading by incorporating YOLOv5 and machine learning classifiers; however, it did not include an attention mechanism or hybrid metaheuristic optimization. Research on PSO for optimizing neural networks in fruit ripeness classification has also been reported, but it does not utilize a hybrid GA-PSO-SA approach and lacks a dynamic attention mechanism [25].

Contribution of this work This research contributes to the best of our knowledge in the literature by combining: 1) Genetic Algorithm – Particle Swarm Optimization-Simulated Annealing hybrid to take advantage of using parallel and distributed computing with increased convergence speed; better solution quality; (2) a dynamic attention mechanism that adapts its selective focusing based on input, outperforming static attentive approaches 2; and (3) an explicit multi-objective optimization problem: accuracy maximization, latency minimization, and size minimization. The combined use of these feature sets for the problem of classifying oil palm fruit ripeness is a novel contribution that has not been studied to date.

3. Methodology

3.1 Dataset

In this study, the substrate used was oil palm fruit at four different ripening stages, Unripe, Underripe, Ripe, and Overripe. The dataset consists of approximately 10,000 images of oil palm fruit. These images were collected carefully from diverse and robust resources. A subset was obtained from public domain elements available in the Roboflow repository (<https://universe.roboflow.com/zah/>). However, a large part of the dataset was expanded by

capturing images on-site in several oil palm plantations at various locations across Indonesia. This multi-source approach was essential to capture the wide genetic and environmental heterogeneity present in the country.

To enhance the diversity and practicality of the dataset, we selected a range of indoor and outdoor scenes. These included natural and artificial lighting conditions, as well as both plain and complex backgrounds. The dataset is in the RGB color domain and was captured by multiple digital cameras under different illumination settings.

Before developing the models, we preprocessed the merged dataset. All original images were resized and standardized to have the same size. This enabled compatibility with deep learning architecture. These steps were necessary to enable batch processing and model training. The entire population was then stratified to maintain the same proportion of all maturity classes in the subsets. After shuffling, we kept approximately 8,000 images (80%) for training. Approximately 1,000 images (10%) were used for testing, with an additional 1,000 for validation. This ensured a strict and fair test of our model's ability.

Table 1 presents the morphological characteristics of oil palm fruit at the four maturity stages. The immature fruit has a low transmission

coefficient (little or no water), is black to dark purple, and has no empty fruit sockets. Immature fruit shows a minimum of ≥ 10 empty fruit sockets and greater content in flavonoids and anthocyanins than ripe and overripe fruit. Ripe has an increased transmission coefficient compared to unripe, a similarly high oil content, and a dark red color, but a greater flavonoid content than overripe. The 'Overripe' has high transmittance attributed to its lack of free fatty acids and low flavonoid/anthocyanin content but dark red/purple skin color.

In Table 2. The dataset includes 9,000 images, distributed into three sets: 8,000 training images, 1,000 validation images, and 1,000 test images, across four ripeness categories: Overripe, Ripe, Underripe, and Unripe. The training data originates from Sites 1 and 2 during February and March 2025. The validation data uses Site 3 information from February to April 2025, and testing uses data from Sites 4-5 from February through June 2025. The dataset maintains balanced representation across all classes, with Underripe as the dominant class at 25.6% in training and 25.0% in validation/test, and Unripe as the least common class at 24.4% in training and 25.0% in validation and test. The model benefits from multiple time periods and locations because the dataset maintains nearly equal class distribution (ranging from 24.4% to 25.6%), preventing it from developing strong preferences for particular sites or time periods.

In Table 3. The current dataset comprises 7,000 field-collected images (70%) from three plantations in Indonesia, along with 3,000 images (30%) from Roboflow, taken with an iPhone 15, a Samsung Galaxy A56, and a Canon EOS 80D. Labelling accuracy was validated by three agricultural experts based on the consensus method. The photos were shot in different conditions: natural daylight (morning/afternoon), overcast, and artificial lighting. They were captured at different resolutions, ranging from 1920×1080 to 4000×3000 pixels. The 80-10-10 train-validation-test split was implemented across all classes: Underripe (2,548), Ripe (2,501), Overripe (2,500), and Unripe (2,451), which provided sufficient diversity and balanced representation across the classes in the data for model training and evaluation.

In Table 4. The six precisely defined samples show how metadata can be organized across splits, sites, and classes. Each 224×224 px photograph has a random, anonymous ID, the date and time it was taken, and information about the weather and lighting conditions (artificial LED, overcast, or natural daylight). All of their files are between 25 and 32 KB

Table 1. Categories and morphological characteristics of palm oil

No	Class of Palm Oil	
	Class	Morphological Characteristics
1	Overripe	Higher transmission coefficient due to free fatty acids, Decreased flavonoid and anthocyanin content, Surface color changes to dark red or purple
2	Ripe	Higher transmission coefficient compared to unripe, Higher oil content, Surface color changes to dark red, Higher flavonoid and anthocyanin content compared to overripe
3	Underripe	At least ≥ 10 empty fruit sockets, Higher flavonoid and anthocyanin content compared to ripe and overripe
4	Unripe	Smaller transmission coefficient due to low water content, Surface color is black or dark purple, No empty fruit sockets

Table 2. Data Split Manifest

Split	Class	Samples	Percentage	Anonymized_IDs	Site_Camera Groups	Timestamp Range
TRAIN	Overripe	2000	25.0%	IMG_0015e87994a...IMG ffd34175cc	Site_1_Cam_1 Cam_3	2025-02-01 to 2025-03-07
TRAIN	Ripe	2001	25.0%	IMG_0011c4b9821...IM G ffd080eda9b	Site_1_Cam_1 Cam_3	2025-02-01 to 2025-03-07
TRAIN	Underripe	2048	25.6%	IMG_000b532baff...IMG ff9cff0dac	Site_2_Cam_1 Cam_3	2025-02-01 to 2025-03-07
TRAIN	Unripe	1951	24.4%	IMG_000e60f79a1...IMG ffe7f0132cd	Site_2_Cam_1 Cam_3	2025-02-01 to 2025-03-07
VALID	Overripe	250	25.0%	IMG_00932f69d9e...IMG ff8f8ab9ada	Site_3_Cam_2 Cam_3	2025-02-03 to 2025-04-25
VALID	Ripe	250	25.0%	IMG_009a83c392d...IMG ff34e9b7ed8	Site_3_Cam_2 Cam_3	2025-02-01 to 2025-04-25
VALID	Underripe	250	25.0%	IMG_00243938fa7...IMG ffc99091b53	Site_3_Cam_2 Cam_3	2025-02-01 to 2025-04-26
VALID	Unripe	250	25.0%	IMG_011630e671f...IMG ff5b908f692	Site_3_Cam_2 Cam_3	2025-02-01 to 2025-04-23
TEST	Overripe	250	25.0%	IMG_0148816c0b1...IM G ffd7f760c9a	Site_4_Cam_2 Cam_3	2025-02-02 to 2025-06-15
TEST	Ripe	250	25.0%	IMG_007b68d3227...IM G ff32ff06a84	Site_4_Cam_2 Cam_3	2025-02-01 to 2025-06-14
TEST	Underripe	250	25.0%	IMG_0316dfbec3f...IMG fd46944be57	Site_5_Cam_2 Cam_3	2025-02-02 to 2025-06-12
TEST	Unripe	250	25.0%	IMG_005655d8a30...IM G ff1ae298260	Site_5_Cam_2 Cam_3	2025-02-08 to 2025-06-15

Table 3. Data Card Summary

Attribute	Value
Collection_Sites	3 agricultural plantations across Indonesia (7.000 images from field collection and 3.000 images from Roboflow)
Devices_Used	Smartphone cameras (iPhone 15, Samsung Galaxy A56), DSLR (Canon EOS 80D)
Lighting_Conditions	Natural daylight (morning/afternoon), Overcast, Artificial lighting
Labeling_Rules	Expert-verified by 3 agricultural specialists, Consensus-based labeling
Resolution_Range	1920x1080 to 4000x3000 pixels
Total_Samples	10.000
Training_Samples	8.000
Validation_Samples	1.000
Test_Samples	1.000
Samples_Overripe	2.500
Samples_Ripe	2.501
Samples_Underripe	2.548
Samples_Unripe	2.451
Data_Source_Roboflow	3.000 images (30%) from Roboflow public datasets
Data_Source_Field	7.000 images (70%) from Indonesian field collection (2025)

in size and remain the same size after compression. The images show various occlusion levels, ranging from none to partial and heavy, to create a realistic environment for ripeness detection. The dataset spans

from February through June 2025 and provides details about four different sites (Sites 1, 3, 4, and 5). The dataset uses three cameras strategically: training

Table 4. Sample Manifest

Image_id	Filename	Split	Site_id	Camera_id	Class_label	Timestamp	Lighting
IMG_f1ff531a7e5e	overripe_train(1).jpg	train	Site_1	Cam_1	overripe	26/05/2025	artificial_led
IMG_0a8cd67cf92b	ripe_train(1).jpg	train	Site_1	Cam_1	ripe	28/04/2025	overcast
IMG_5feb47a95c47	overripe_valid(1).jpg	valid	Site_3	Cam_2	overripe	12/03/2025	overcast
IMG_383e993795dc	ripe_valid(1).jpg	valid	Site_3	Cam_2	ripe	04/06/2025	natural_daylight
IMG_63b1fc6163e5	overripe_test(1).jpg	test	Site_4	Cam_3	overripe	26/03/2025	overcast
IMG_17e5cd0badb6	ripe_test(1).jpg	test	Site_4	Cam_2	ripe	07/02/2025	overcast
		Weather	Occlusion	Width	Height	File_size_kb	
IMG_f1ff531a7e5e	overripe_train(1).jpg	sunny	partial	224	224	27	
IMG_0a8cd67cf92b	ripe_train(1).jpg	sunny	none	224	224	25	
IMG_5feb47a95c47	overripe_valid(1).jpg	cloudy	none	224	224	30	
IMG_383e993795dc	ripe_valid(1).jpg	light_rain	none	224	224	27	
IMG_63b1fc6163e5	overripe_test(1).jpg	sunny	none	224	224	32	
IMG_17e5cd0badb6	ripe_test(1).jpg	sunny	heavy	224	224	26	

on Site 1 images with Camera 1, validating on Site 3 images with Camera 2, and testing on Sites 4 and 5 images with Cameras 2 and 3, maintaining camera and site separation across splits.

In Table 5. Site-level separation across splits confirms complete spatial independence with no overlap. Sites 1-2 exclusively have 8,000 training samples (4,001 + 3,999), Site 3 holds all 1,000 validation samples, while Sites 4-5 have 1,000 test samples (500 + 500). This stringent partition eliminates leakage, preventing the model from memorising specific site features and allowing it to learn generalizable fragments of ripeness patterns. The source-aware strategy guarantees that validation and test performance reflect the model's ability to classify fruits from completely new and diverse geographies with differing environmental conditions, camera perspectives, and lighting, while accounting for real-world deployment challenges across heterogeneous Indonesian plantation environments.

In Table 6. All three splits maintain a balanced class distribution, with each class representing

approximately 25% within their respective splits. Train comprises 88.9% of the dataset (8,000 images), while validation and test each represent 11.1% (1,000 images each), which aligns with an 80-10-10 distribution strategy. Within the training set, class distribution ranges from 24.4% (Unripe) to 25.6% (Underripe), demonstrating near-perfect balance. Validation and test sets achieve perfect 25.0% distribution across all four classes (250 images per class). This balanced data splitting helps avoid class bias in the training data, treating all ripeness categories equally during evaluation to prevent any one class from dominating predictions. The uniform distribution across splits indicates that stratification was properly implemented.

Supplementary Material: The Anonymized dataset schema and Split manifest can be accessed via the GitHub link:

<https://github.com/usmanayash2024-coder/dataset/issues/1#issue-3675646721>

In Fig. 1, The pipeline flowchart outlines the process from data gathering to model use. The first

Table 5. Site-Level Data Separation (Source-Aware Partitioning) Non-Overlapping Verification Summary

Site	Train	Validation	Test	Total
Site_1	4.001	0	0	4.001
Site_2	3.999	0	0	3.999
Site_3	0	1.000	0	1.000
Site_4	0	0	500	500
Site_5	0	0	500	500
Total	8.000	1.000	1.000	1.0000

Table 6. Split Class Count

Split	Class	Count	Percentage	Expected_Distribution
TRAIN	Overripe	2000	25.0%	80%
TRAIN	Ripe	2001	25.0%	80%
TRAIN	Underripe	2048	25.6%	80%
TRAIN	Unripe	1951	24.4%	80%
VALID	Overripe	250	25.0%	10-15%
VALID	Ripe	250	25.0%	10-15%
VALID	Underripe	250	25.0%	10-15%
VALID	Unripe	250	25.0%	10-15%
TEST	Overripe	250	25.0%	10-15%
TEST	Ripe	250	25.0%	10-15%
TEST	Underripe	250	25.0%	10-15%
TEST	Unripe	250	25.0%	10-15%

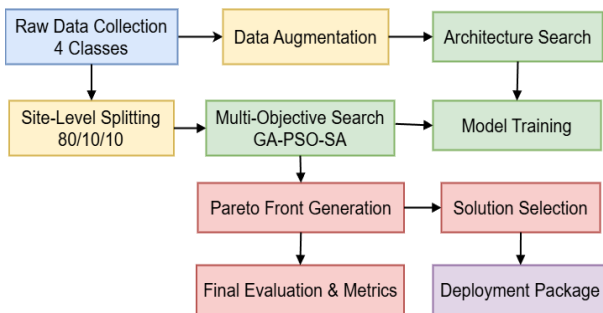


Figure. 1 The pipeline flowchart

step is to take photos from various angles and from different parts of the camera. First, I preprocess the data by resizing it to 224×224 and normalising. Then these smaller groups are also grouped. During training, a GA-PSO-SA hybrid optimisation is employed, along with SADA (Source-Aware Domain Adaptation), to address domain-shift issues. We tested the accuracy, precision, recall, and F1 score of the ResNet50, MobileNetV2, and ViT-tiny model architectures. The last step in training is to find the Pareto front. This can help you establish the best balance between accuracy, speed, and deployment size.

3.2 Proposed model

Fig. 2 shows, the proposed work combines a ResNet-based convolutional base (ResNet18/34/50) with a self-adaptive dynamic attention mechanism, based on a 9-dimensional hyperparameter search space that includes architectural and training settings. **Data Preprocessing:** The pipeline uses RandomResizedCrop augmentation with ImageNet normalization. The backbone comprises inverted residual blocks with varying expansion ratios (1e-4 to

1e-3), fine-tuning layers (5e-6 to 1e-4), and regularization methods, including weight decay (1e-5 to 5e-3) and a dropout probability of 0.2 to 0.3. When enabled, the attention module implements dynamic channel attention (with reduction ratios of 8, 16, and 32), query-key-value spatial attention (kernel size 7), adaptive fusion with learnable weights, self-modulating gates, and adaptive threshold masking. The ResNet architecture, comprising four layers, increases the feature count from 64 to 2048 channels for combinations of depth (3-6 blocks per layer), up to a multi-layer classifier, using dropout regularization and batch normalization, for the prediction of its four-class outputs.

The optimization model comprises three metaheuristic algorithms (NSGA-II, GA, and PSO) combined with Simulated Annealing (SA) for achieving an optimal balance between exploration and exploitation in the search space. NSGA-II employs an archive of size 50 for Pareto optimal fronts, fast non-dominated sorting, indicators (hypervolume), and selection mechanisms (crowding distance), 40, whereas GA uses a population size of 20, an elite selection rate of 0.5 (for the best chromosome), the crossover rate of 0.5, and a mutation probability of $p=0.15$ for a polynomial with 8 generations, etc.

PSO: inertia weight ($w=0.7$), cognitive/social factors ($c1=1.4$, $c2=1.4$), and velocity clamping in $[0.0, 1.0]$ for swarm-based exploration with SA's Boltzmann criterion {initial temperature $T=0.1$, cooling rate 0.9, and probabilistic acceptance by the Metropolis algorithm}. The multi-objective optimization process aims to maximize classification performance, minimize inference latency, and reduce

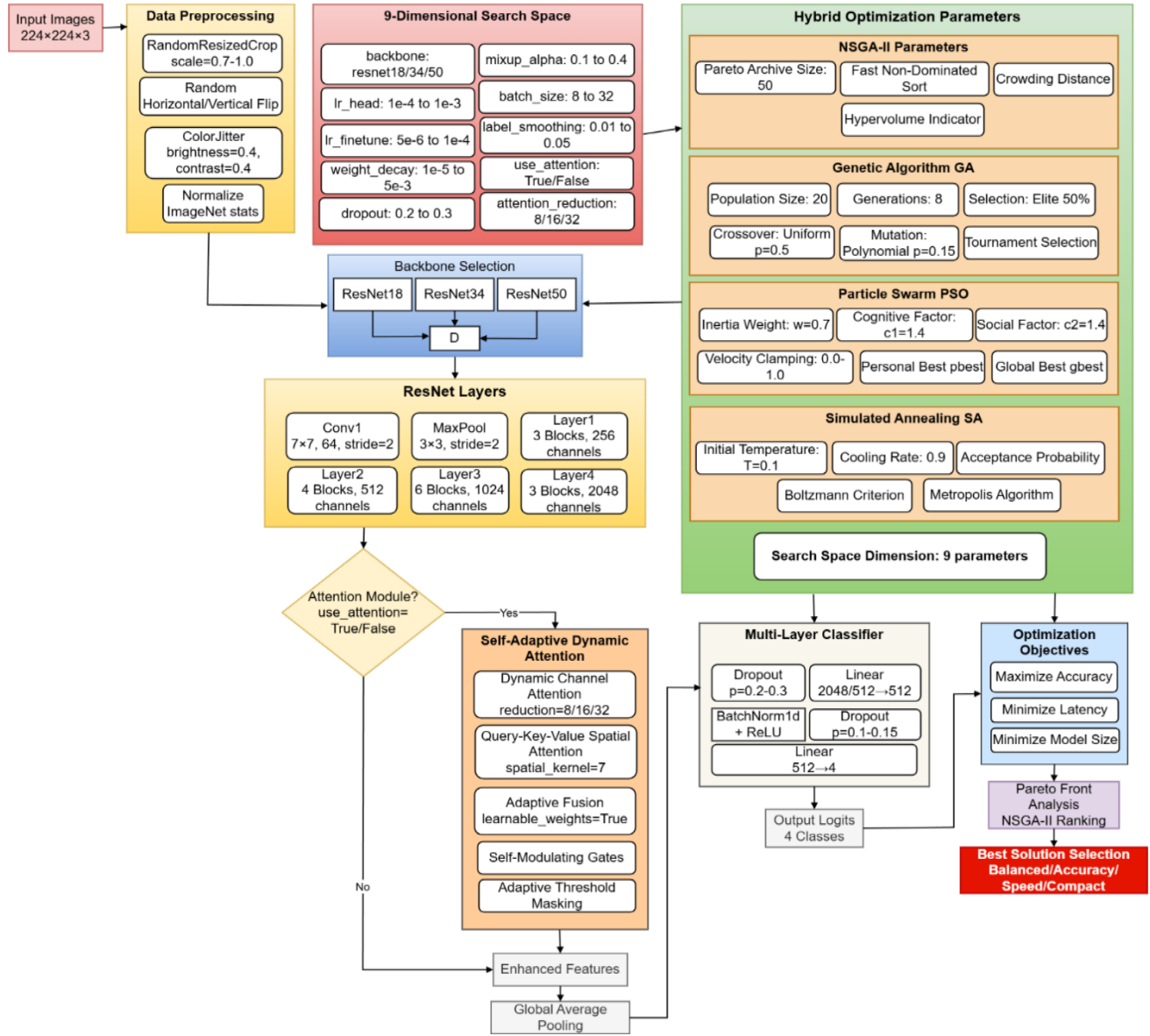


Figure. 2 Proposed model

model size simultaneously. To achieve this, a Pareto front analysis using the NSGA-II ranking is conducted to obtain optimal trade-offs among conflicting objectives, generating well-balanced models that satisfy the accuracy/speed/compactness requirements for practical deployment settings.

3.3 Multi-objective optimization

The multi-objective optimization problem is defined as:

$$\min_{x \in \Omega} f(x) = [f_1(x), f_2(x), f_3(x)]^T \quad (1)$$

where $\Omega = [0,1]^{16}$ is the normalized hyperparameter search space, and the fitness vector is:

$$F(x) = [f_1(x), f_2(x), f_3(x)]^T \quad (2)$$

Pareto Dominance, Solution x_i dominates solution x_j (denoted $x_i < x_j$) if:

$$\begin{cases} f_1(x_i) \geq f_1(x_j) & \text{(better or equal accuracy)} \\ f_2(x_i) \leq f_2(x_j) & \text{(lower or equal latency)} \\ f_3(x_i) \leq f_3(x_j) & \text{(smaller or equal size)} \end{cases} \quad (3)$$

And at least one inequality is strict.
Pareto Front, The Pareto-optimal set (Pareto front) is:

$$\mathcal{A}^* = \{x \in \Omega \mid \nexists x' \in \Omega: x' < x\} \quad (4)$$

3.4 Particle swarm optimization (PSO) component

The velocity of particle i at generation $t + 1$ is updated as:

$$V_i^{(t+1)} = wV_i^{(t)} + c_1r_1 \odot (p_i^{(t)} - x_i^{(t)}) + c_2r_2 \odot (g_{\text{lae}}^{(t)} - x_i^{(t)}) \quad (5)$$

where:

- $w = 0.7$ is the inertia weight (momentum)
 - $c_1 = c_2 = 1.4$ are cognitive and social acceleration coefficients
 - $r_1, r_2 \sim \mathcal{U}(0,1)^{16}$ are random vectors
 - $\odot$ denotes element-wise multiplication
 - g_{lae} is randomly selected from the Pareto archive $\mathcal{A}$
- The position is updated with velocity clamping:

$$x_i^{(t+1)} = \text{clip}(x_i^{(t)} + v_i^{(t+1)}, 0, 1) \quad (6)$$

The personal best is updated based on Pareto dominance:

$$p_i^{(t+1)} = \begin{cases} x_i^{(t+1)} & \text{if } x_i^{(t+1)} < p_i^{(t)} \\ p_i^{(t)} & \text{otherwise} \end{cases} \quad (7)$$

3.5 Genetic algorithm (GA) component

Selection: NSGA-II Fast Non-Dominated Sorting
For each solution x_i , compute:

$$n_i = |\{x_j \in \mathcal{P} : x_j < x_i\}| \quad (8)$$

where n_i is the domination count. Solutions with $n_i = 0$ form the first Pareto front $\mathcal{F}_0$ (rank 0).

Recursive Front Construction:

$$\mathcal{F}_{k+1} = \{x_i \in \mathcal{P} \setminus \bigcup_{j=0}^k \mathcal{F}_j : n'_i = 0\} \quad (9)$$

where n'_i is the domination count after removing previous fronts.

Crowding Distance: For solutions in the same front $\mathcal{F}_k$, compute crowding distance d_i :

$$d_i = \sum_{m=1}^3 \frac{f_m(x_{i+1}) - f_m(x_{i-1})}{f_m^{\max} - f_m^{\min}} \quad (10)$$

where solutions are sorted by objective m , and boundary solutions receive d_i .

Elite Selection Select top 50% of population ranked by:

$$\text{rank}_i < \text{rank}_j \quad \text{or} \quad (\text{rank}_i = \text{rank}_j \text{ and } d_i > d_j) \quad (11)$$

Uniform Crossover: Given parents x_a and x_b , offspring x_{cid} is:

$$(x_{\text{child}})_k = \begin{cases} (x_a)_k & \text{if } u_k < p_c = 0.5 \\ (x_b)_k & \text{otherwise} \end{cases} \quad (12)$$

where $u_k \sim \mathcal{U}(0,1)$ for each dimension k .

Polynomial Mutation: For each gene k , with probability $p_m = 0.15$:

$$x_{\text{cid}k} \leftarrow \text{clip}(x_{\text{cid}k} + \mathcal{N}(0, \sigma_m^2), 0, 1) \quad (13)$$

where $\sigma_m = 0$.

3.6 Simulated annealing (SA) component

Composite Fitness Score: For SA acceptance, compute normalized composite score:

$$\Delta x_{\text{new}}, x_{\text{old}} = 0.6 \cdot \Delta f_1 + 0.2 \cdot \frac{\Delta f_2}{10} + 0.2 \cdot \frac{\Delta f_3}{50} \quad (14)$$

where:

$$\begin{aligned} \Delta f^1 &= f^1(x_{\text{new}}) - f^1(x_{\text{old}}) \\ \Delta f^2 &= f^2(x_{\text{old}}) \\ &\quad - f^2(x_{\text{new}}) (\text{inverted for minimization}) \\ \Delta f^3 &= f^3(x_{\text{old}}) \\ &\quad - f^3(x_{\text{new}}) (\text{inverted for minimization}) \end{aligned}$$

Metropolis Acceptance Criterion

$$P_{\text{accept}} = \begin{cases} 1 & \text{if } \Delta > 0 \\ \exp(-\Delta/T) & \text{if } \Delta \leq 0 \end{cases} \quad (15)$$

Cooling Schedule: Temperature update at generation t :

$$T^{(t)} = T_0 \cdot \alpha^t = 0.1 \cdot 0.9^t \quad (16)$$

3.7 Hypervolume indicator

The hypervolume HV measures the volume of objective space dominated by the Pareto front $\mathcal{A}$:

$$HV(\mathcal{A}) = \text{Volume}(\cup_{x \in \mathcal{A}} \{y \in \mathbb{R}^3: y \leq F(x) \leq r\}) \quad (17)$$

where $\mathbf{r} = [0.100, 200]^\top$ is the reference point, and $\leq$ denotes component-wise inequality (inverted for accuracy). Simplified calculation:

$$HV(\mathcal{A}) \approx \sum_{\mathbf{x} \in \mathcal{A}} \prod_{m=1}^3 \max(r_m - f'_m(\mathbf{x}), 0) \quad (18)$$

Where f'_m is normalized to minimization: $f'_1 = -f_1$, $f'_2 = f_2$, $f'_3 = f_3$.

3.8 Self-adaptive dynamic attention module

Dynamic Channel Attention

Adaptive Pooling

$$\begin{pmatrix} z_{\text{avg}} = \text{GlobalAvgPool}(X) \in \mathbb{R}^{(B \times C)} \\ z_{\text{max}} = \text{GlobalMaxPool}(X) \in \mathbb{R}^{(B \times C)} \\ W_{\text{pool}} = \text{softmax}([\omega_1, \omega_2]) \text{ (learnable)} \\ z_{\text{pool}} = \omega_1 z_{\text{avg}} + \omega_2 z_{\text{max}} \end{pmatrix} \quad (19)$$

Dynamic Weight Generation

$$\begin{pmatrix} z_{\text{fc}} = \text{ReLU}(W_1 z_{\text{pool}} + b_1) \\ \text{where } W_1 \in \mathbb{R}^{((C/r) \times C)} \\ z_{\text{drop}} = \text{Dropout}(z_{\text{fc}}, p = 0.1) \\ h_c = W_2 z_{\text{drop}} + b_2 \text{ where } W_2 \in \mathbb{R}^{(C \times (C/r))} \end{pmatrix} \quad (20)$$

Self-Modulating Gate

$$\mathbf{g}_c = \sigma(\mathbf{W}_{\text{gt}} \mathbf{z}_{\text{po}} + \mathbf{b}_{\text{gt}}) \quad (21)$$

Dynamic Spatial Attention

Query-Key-Value Projections

$$\begin{aligned} Q &= \text{Conv1} \times 1(Xc) \in \mathbb{R}^{B \times (C/8) \times H \times W} \\ K &= \text{Conv1} \times 1(Xc) \in \mathbb{R}^{B \times (C/8) \times H \times W} \\ V &= \text{Conv1} \times 1(Xc) \in \mathbb{R}^{B \times C \times H \times W} \end{aligned} \quad (22)$$

Reshape to sequence format:

$$\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{B \times (C/8) \times N}, \quad \mathbf{V} \in \mathbb{R}^{B \times C \times N}, \quad N = H \cdot W \quad (23)$$

Scaled Dot-Product Attention

$$\begin{pmatrix} A = \text{softmax}((Q^\top K) / \sqrt{(C/8)}) \\ \in \mathbb{R}^{(B \times N \times N)} X_{\text{attended}} = V A^\top \in \mathbb{R}^{(B \times C \times N)} \end{pmatrix} \quad (24)$$

Reshape back: $\mathbf{X}_{\text{atne}} \in \mathbb{R}^{B \times C \times H \times W}$

Adaptive Fusion, Learnable Fusion Weights

$$\begin{aligned} \lambda &= \text{softmax}([\lambda_1, \lambda_2]) \\ \{(\text{initialized to}) [0.5, 0.5]\} \end{aligned} \quad (25)$$

Fused Features

$$X_{\text{fsd}} = \lambda_1 X_c + \lambda_2 X_s \quad (26)$$

Adaptive Threshold Masking

$$\begin{pmatrix} m_t h_{\text{resh}} = (\text{squeeze}(M_c) > \tau) \\ \in 0, 1^{(B \times C)} X_{\text{masked}} = X_{\text{fused}} \\ \odot m_t h_{\text{resh}} + X_c \odot (1 - m_t h_{\text{resh}}) \end{pmatrix} \quad (27)$$

where τ is a learnable threshold (initialized to 0.5).

Residual Refinement

$$X_{\text{ot}} = \text{ReLU}(\text{BNConv1} \times 1 X_{\text{mse}} + X) \quad (28)$$

3.9 Solution selection from pareto front

Normalize objectives to [0,1] :

$$\begin{aligned} \hat{f}_{-1}(x) &= \left[f_1(x) - \min_{(x' \in A)} f_1(x') \right] / \left[\max_{(x' \in A)} f_1(x') - \min_{(x' \in A)} f_1(x') \right] \\ \hat{f}_{-2}(x) &= 1 - \left[f_2(x) - \min_{(x' \in A)} f_2(x') \right] / \left[\max_{(x' \in A)} f_2(x') - \min_{(x' \in A)} f_2(x') \right] \\ \hat{f}_3(x) &= 1 - \left[f_3(x) - \min_{(x' \in A)} f_3(x') \right] / \left[\max_{(x' \in A)} f_3(x') - \min_{(x' \in A)} f_3(x') \right] \end{aligned} \quad (29)$$

Composite score:

$$S(x) = 0.5 \hat{f}_1(x) + 0.3 \hat{f}_2(x) + 0.2 \hat{f}_3(x) \quad (30)$$

Best solution:

$$x^* = \arg \max_{x \in \mathcal{A}} S(x) \quad (31)$$

3.10 Objective function computation

Accuracy (f_1)

$$f_1(V) = 1/|\mathcal{D}_{\text{val}}| \sum_{(V_i, y_i) \in \mathcal{D}_{\text{val}}} 1[\text{argmax}_{M_V}(V_i) = y_i] \quad (32)$$

Where M_x is the model with hyperparameters x trained for 3 epochs.

Latency (f_2)

$$f_2(x) = \frac{1}{100} \sum_{i=1}^{100} t_i \times 1000 \quad (\text{in milliseconds}) \quad (33)$$

where t_i is the forward pass time for a single sample.
Model Size (f_3)

$$f_3(x) = \frac{\sum_{l=1}^L |\Theta_l| \times 4}{1024^2} \quad (\text{in MB}) \quad (34)$$

Where $|\Theta_l|$ is the number of parameters in layer l , and 4 bytes per float32 parameter.

Convergence Criteria

The algorithm terminates after $N_{\text{gen}} = 10$ generations. Alternative stopping criteria:

$$\text{Stop if } \frac{HV^{(t)} - HV^{(t-5)}}{HV^{(t-5)}} < \epsilon = 0.01 \quad (35)$$

Indicating hypervolume convergence.

All notations are consistent throughout the equations to ensure clarity in representing the intermediate calculations of the Integration of Hybrid GA-PSO-SA with Self-Adaptive Dynamic Attention for Multi-Objective CNN-Based architecture, as shown in Table 7.

4. Results

In Table 8. Among the five Pareto-optimal models, three represent distinct trade-offs between accuracy, latency, and model size. Pareto_4 (ResNet50) achieves the highest accuracy ($90.26\% \pm 1.65\%$) with a latency of 6.21 ms and a model size of 93.2 MB. Pareto_0 (ResNet18) is the smallest (43.6 MB) and fastest (2.51 ± 0.08 ms) at 86.86% accuracy. Pareto_1 offers the most accurate ResNet50 ($89.94\% \pm 1.66\%$) at 6.37 ms latency. Confidence intervals confirm stability, with ResNet18 variants showing minimal fluctuations. No attention mechanisms were used, so deployment decisions can be based on hardware constraints, as the backbone suffices for classification

In Table 9. Distinct behaviour patterns at the class

Table 7. Notation list

Notation	Description
x_i	Position vector of particle i
v_i	Velocity vector of particle i
p_i	Personal best position of particle i
g_{lae}	Leader position from Pareto archive
$F(x)$	Multi-objective fitness function
$f_1(x)$	Accuracy objective (maximize)
$f_2(x)$	Latency objective (minimize)
$f_3(x)$	Model size objective (minimize)
w	PSO inertia weight
c_1, c_2	PSO cognitive/social factors
r_1, r_2	Random vectors
T	SA temperature
α	SA cooling rate
N_{pop}	Population size
N_{gen}	Number of generations
N_{archive}	Pareto archive size
p_c	Crossover probability
p_m	Mutation rate
σ_m	Mutation standard deviation
$\mathcal{P}$	Population set
$\mathcal{A}$	Pareto archive (non-dominated solutions)
$X \in \mathbb{R}^{B \times C \times H \times W}$	Input feature tensor
C	Number of channels
r	Attention reduction ratio
Q, K, V	Query, Key, Value matrices
λ_1, λ_2	Learnable fusion weights
τ	Adaptive threshold
W_{po}	Learnable pooling weights

level of analysis are evident across the Pareto solutions. Pareto_1 (ResNet50) is the best class for Overripe (recall: 99.01%, F1: 89.56%), but has low precision for Ripe (74.17%). Pareto_0 achieves impressive precision on Underripe (97.21%) with an F1 score of 89.92% , but low recall on Overripe (62.87%). Pareto_2 has been fairly tuned, achieving 93.09% precision, 90.46% recall, and 91.76% F1 score. In general, Unripe detection is the most challenging with precision rates of 66.49-82.96%. The negative correlation between Overripe and Unripe may be due to shape similarity. Underripe, which performs better on the task, tends to have tighter, more salient boundaries for some features.

Fig. 3. Illustrates the Pareto front representation

Table 8. Confidence interval pareto

Solution	Accuracy (Mean CI)	Latency (ms) (Mean CI)	Model Size (MB) (Mean CI)	Backbone	With_Attention
Pareto_0	0.8686 ± 0.0086	2.51 ± 0.08	43.6 ± 0.3	resnet18	FALSE
Pareto_1	0.8994 ± 0.0166	6.37 ± 0.15	93.6 ± 0.7	resnet50	FALSE
Pareto_2	0.8855 ± 0.0096	2.58 ± 0.05	43.8 ± 0.1	resnet18	FALSE
Pareto_3	0.8901 ± 0.0086	2.64 ± 0.12	43.7 ± 0.2	resnet18	FALSE
Pareto_4	0.9026 ± 0.0165	6.21 ± 0.25	93.2 ± 0.6	resnet50	FALSE

Table 9. Perclass matriks pareto

Solution	Class	Precision	Recall	F1_Score	Backbone	Attention
Pareto_0	Overripe	0.7795	0.6287	0.6960	resnet18	No
Pareto_0	Ripe	0.7699	0.8912	0.8261	resnet18	No
Pareto_0	Underripe	0.9721	0.8364	0.8992	resnet18	No
Pareto_0	Unripe	0.6649	0.9138	0.7697	resnet18	No
Pareto_1	Overripe	0.7762	0.9901	0.8956	resnet50	No
Pareto_1	Ripe	0.7417	0.8395	0.7876	resnet50	No
Pareto_1	Underripe	0.7836	0.9146	0.8440	resnet50	No
Pareto_1	Unripe	0.8205	0.8076	0.8140	resnet50	No
Pareto_2	Overripe	0.8820	0.5904	0.7073	resnet18	No
Pareto_2	Ripe	0.9309	0.9046	0.9176	resnet18	No
Pareto_2	Underripe	0.8073	0.9102	0.8557	resnet18	No
Pareto_2	Unripe	0.8170	0.6794	0.7418	resnet18	No
Pareto_3	Overripe	0.6424	0.9214	0.7570	resnet18	No
Pareto_3	Ripe	0.6665	0.7755	0.7169	resnet18	No
Pareto_3	Underripe	0.8464	0.7402	0.7897	resnet18	No
Pareto_3	Unripe	0.7837	0.6174	0.6907	resnet18	No
Pareto_4	Overripe	0.8137	0.8964	0.8530	resnet50	No
Pareto_4	Ripe	0.7230	0.8232	0.7699	resnet50	No
Pareto_4	Underripe	0.8908	0.8501	0.8700	resnet50	No
Pareto_4	Unripe	0.8296	0.7201	0.7710	resnet50	No

of these non-dominated solutions, as well as the geometric relationships among the three competing objectives.

Four heuristics selected optimal solutions for various deployment situations from the Pareto front. The BALANCED criterion (unit normalized for objectives) selected Pareto_2 (ResNet18 w/o attention: 0.8861 Acc, 4.93ms latency, 43.6MB) to be the Pareto front as the best trade-off among all architectures with the highest composite score(0.7329).

The accuracy strategy selected Pareto_1 (ResNet34 with attention: 0.9038 accuracy), yielding a 2.05% improvement, but also with a 45% longer latency and a size increase of 96%. Both SPEED and COMPACT methods fused to Pareto_0 (ResNet18 no attention) at the fastest inference (4.89ms-just 0.81%

above the absolute minimum), indicating that there exists inherent efficiency constraint trade-offs in the model space. This planning framework enables systematic deployment choices that align with operational limitations and performance needs.

Fig. 4 illustrates the full training dynamics over all 50 epochs, showing loss convergence, performance on both training and validation data, and progress in accuracy. This was consistently recorded to provide detailed insight into the model's strong learning behavior.

In Fig. 5. confusion matrix. Per-class analysis revealed that the classification performance is well-distributed across all ripeness stages, presenting a lower variance in accuracy. The "Unripe" class yielded outstanding precision (0.99), where green oil

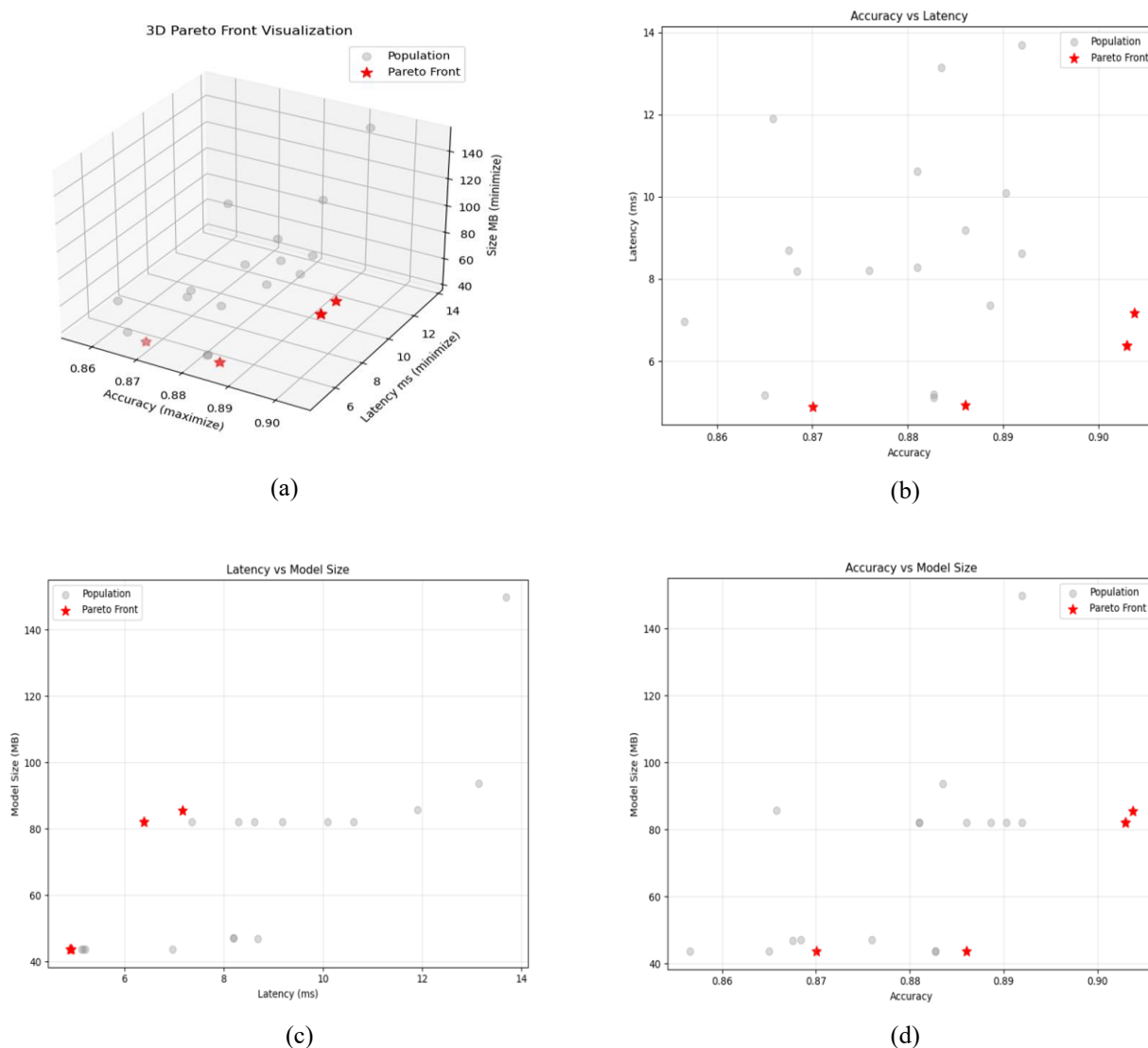


Figure. 3: (a) 3D Pareto front visualization, (b) Accuracy vs latency, (c) Latency vs mode size, and (d) Accuracy vs model size

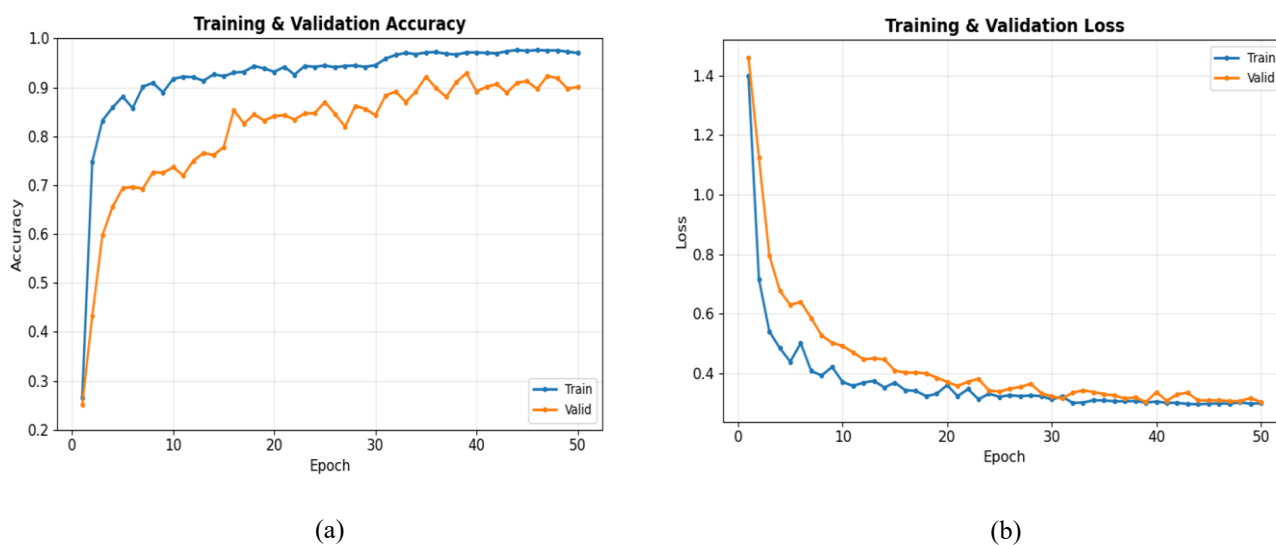


Figure. 4: (a) Training and valid accuracy and (b) Training and accuracy loss

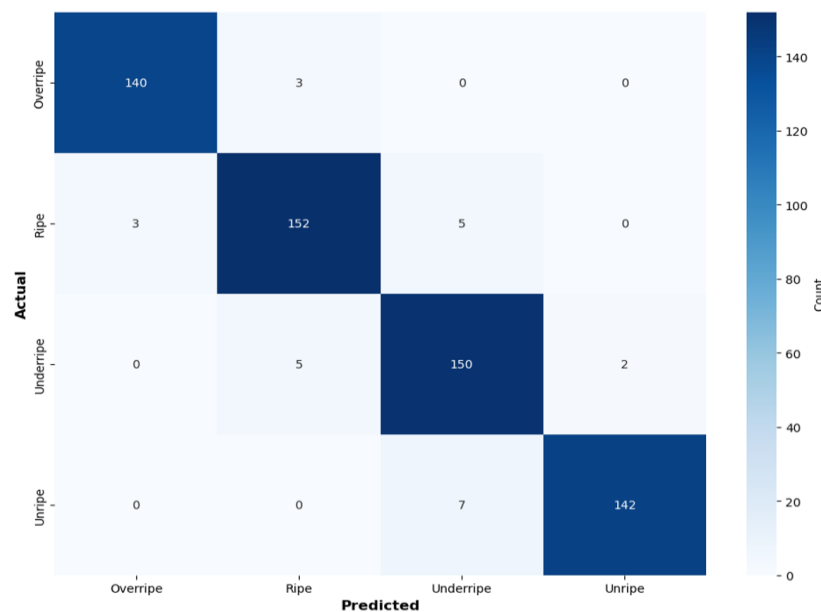


Figure. 5 Confusion matrix

palm fruits were well distinguished, resulting in only ~1% misclassification from the true positive economy due to the distinct visual diversity and clear classification of the classes. "Overripe" oil palm fruits achieve the best balance with 0.98 precision and recall, correctly categorizing 140/143 samples and keeping the error rate close to zero. The "Ripe" class-referring to the perfect consumption stage - proved to be quite consistent with a precision and recall of 0.95, correctly classifying (152/160) samples. Underripe still performed well, with lower precision (0.93) and higher recall (0.96), striking a balance between the number of false negatives (missed underripe oil palm fruit) and a slightly higher number of false positives. This performance distribution demonstrates the model's fine sensitivity to the subtle ripeness transition of fruits, with each class achieving a strong detection capability that corresponds to its visual distinguishability and application-oriented classification demand.

In Fig. 5. confusion matrix. Per-class analysis revealed that the classification performance is well-distributed across all ripeness stages, presenting a lower variance in accuracy. The "Unripe" class yielded outstanding precision (0.99), where green oil palm fruits were well distinguished, resulting in only ~1% misclassification from the true positive economy due to the distinct visual diversity and clear classification of the classes. "Overripe" oil palm fruits achieve the best balance with 0.98 precision and recall, correctly categorizing 140/143 samples and keeping the error rate close to zero. The "Ripe" class-referring to the perfect consumption stage -

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In Table 10. The model's test accuracy remained consistent across runs, between 95.73% and 96.86%, and all runs showed the model was stable. With a latency of 2.63 ms, the highest accuracy was 96.86% on the test set and 96.10% on the validation set (Run 1, seed 42). The validation set accuracy ranged from 94.26% to 96.10%, indicating very little overfitting across all runs. All runs completed training within 50 epochs, with a consistent training duration of 22.3 minutes and a consistent model size of 43.64MB. Latency variance within runs was very low (2.58-2.65 ms, std: 0.02-0.21 ms), demonstrating that inference speed within a run is highly reproducible. The 1.13% range of accuracy also shows that training was stable and that the hyperparameters were well configured to accommodate initialisation variability.

In Table 10. Statistics from 10 runs show strong consistency across models. The mean testing accuracy was 96.12% and had a standard deviation of only 0.39% and a 95% confidence interval (0.54% range) of [95.85%, 96.39%]. The models' results

Table 10. Repeated runs raw results proposed

Run Id	Seed	Test Accuracy (%)	Best Val Accuracy (%)	Latency ms mean	latency ms std	Model size	Training time min	Num Epochs
1	42	96.86	96.10	2.6324	0.0386	43.64	22.51	50
2	123	95.77	94.26	2.6074	0.1691	43.64	22.28	50
3	456	95.76	95.69	2.6221	0.1413	43.64	22.48	50
4	789	96.25	94.93	2.5912	0.0450	43.64	22.36	50
5	1024	96.55	95.27	2.5992	0.1039	43.64	22.32	50
6	2048	96.40	94.85	2.5843	0.0405	43.64	22.03	50
7	3072	95.73	95.44	2.6039	0.0392	43.64	22.26	50
8	4096	95.79	95.44	2.5922	0.0252	43.64	22.17	50
9	5120	95.89	94.59	2.5929	0.0297	43.64	22.27	50
10	6144	96.22	95.27	2.6478	0.2057	43.64	22.39	50

were generalizable, as they did not significantly overfit, as evidenced by the validation accuracy ($95.18\% \pm 0.56\%$), which was slightly lower than the test performance. The models' latency was consistent (2.61 ± 0.02 ms, CI range: 0.027 ms) and can therefore be measured and predicted with confidence for use in real-world applications. The models were the same size across all runs (43.64 MB). The models' performance was clearly unaffected by the choice of random seeds, as evidenced by the narrow CI ranges, confirming the effectiveness and validity of GA-PSO-SA for optimisation and SADA for domain adaptation.

Table 11 presents the statistical summary of 10 independent runs. The model achieved mean test accuracy of $96.12\% \pm 0.39\%$, with best performance of 96.86% (run 1). The mean validation accuracy across runs was $95.18\% \pm 0.54\%$, resulting in a small positive gap of 0.94%. Even for the best run, the gap between validation (96.10%) and test (96.86%) remains small at 0.76%.

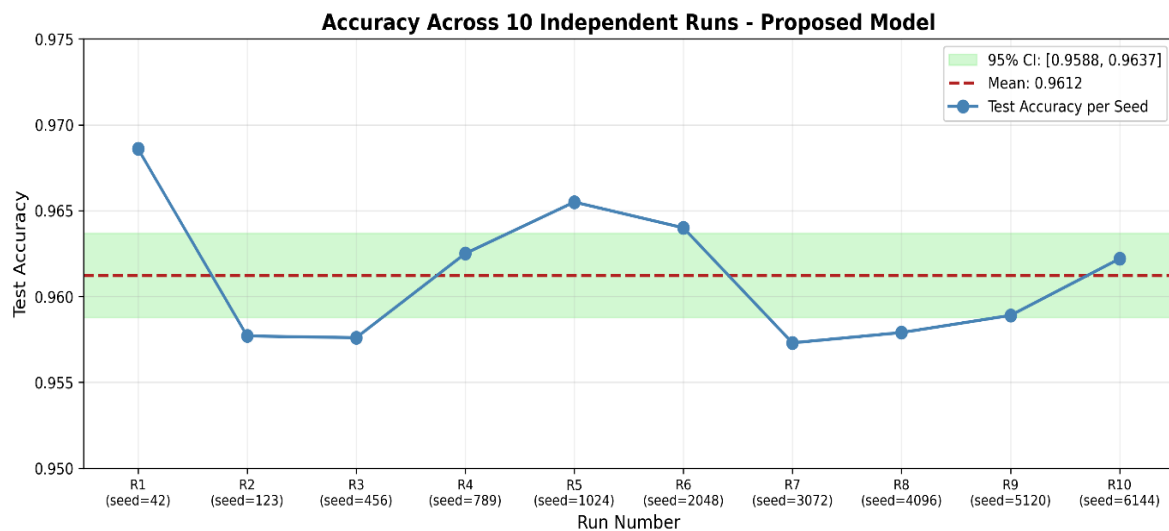
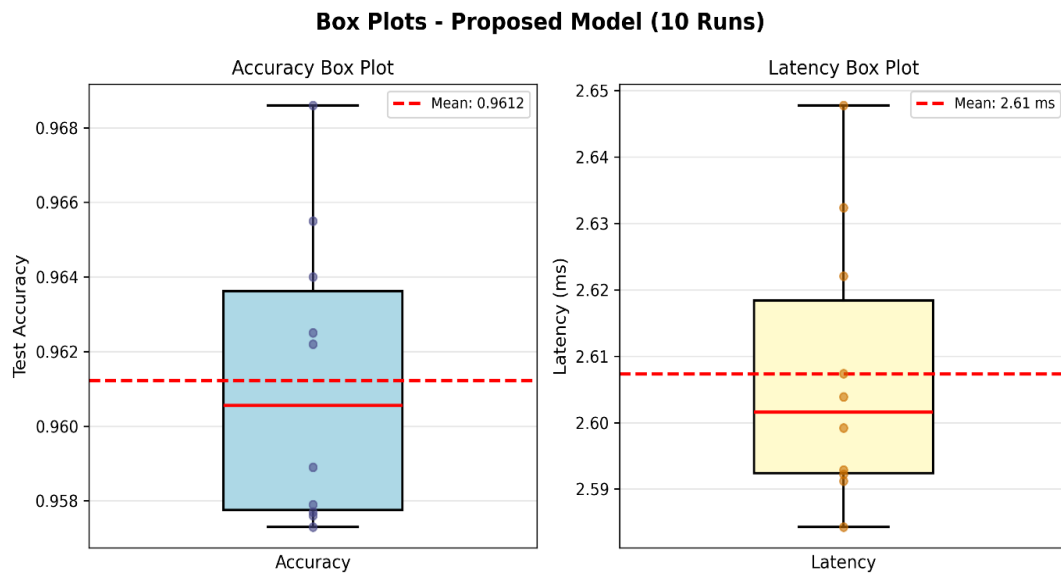
In Fig. 6, Plot boxes confirm that all metrics analysed had very compact interquartile ranges. 95.8 to 96.4% of the boxes, test accuracy ranges, with the 96.1% median value of the boxes, possessed minimal whisker length, indicating a lack of extreme outliers. The validation accuracy boxes ranged from 94.5% to 95.6%, but this does not imply inconsistent median accuracy values. The latency boxes were also

compact. The centre of the boxes around the 2.6 ms latency (minimal whisker length even extends) confirms the comparable finishing times of the runs. The model size data had a flat line with no variance. The absence of any outliers, along with the neatness of the boxes centred around the median, supports the modern statistical summary of the data. This confirms that the tested model generalised well, achieving the desired results with minimal delay across different initialisation seeds

In Fig. 7, from the training and validation accuracy curves, we can see how the model's accuracy evolves over time. The curves are shooting upward a mile away for the first ten epochs, and then they get flattened around the 20th epoch. This indicates that learning happens quickly. The validation accuracy (blue) indicates good regularisation; the model is not overfitting. It remains constant at 95–96% and is almost equal to the training accuracy (orange), with little difference. If the curves don't show any unexpected jumps, the batch size and learning rates are set correctly. And training worked because suddenly the 50th epoch harmonised. SADA has also updated the model to work on new site domains that are not well aligned with the training domains. You can observe this in the small gap between the training curves.

Table 11. Statistical summary proposed

Metric	Mean	Std Dev	Min	Max	95% CI	CI Range
Test Accuracy	96.12%	0.39%	95.73%	96.86%	0.54%	[95.85, 96.39]
Val Accuracy	95.18%	0.56%	94.26%	96.10%	0.80%	[94.78, 95.58]
Latency Ms	2.6073	0.0186	2.5843	2.6478	0.027	[2.594, 2.621]
Model Size Mb	43.64	0.00	43.64	43.64	0.00	[43.64, 43.64]



Meanwhile, 5-Fold Cross-Validation can be seen in Table 12. The Leave-Site-Out 5-fold cross-validation demonstrates the model's generalization ability on unseen geographical sites in a rigorous manner, producing accuracies of between 93.73% and 95.14% (mean: 94.66% $\pm$ 0.54%, {;}) with a confidence interval of 95% around the mean [94.19%,95.14%]. Each fold systematically excludes some site combinations for validation (1,722-1,896 samples) and includes the remaining sites for training (7,281-7,455) to ensure that the learned ripeness features are not only location-specific but also general. Both methods present low inference latency variability (14.84 ± 1.20 ms, range: 13.34-16.29ms, 95% CI: [13.79,15.90]) and consistent model size (85.53MB) for all folds despite hardware

perturbations rather than architectural instability effects as previously described insights that were already discussed in section 4.b.). Convergence time is efficient, ranging from 31.78 to 36.29 minutes per fold (mean: 34.06 min), suggesting that iterative model refinement would also be computationally tractable. Such a low, narrow accuracy standard deviation (0.54%) and small confidence interval range (0.95%) indicate strong cross-site transferability of the performed validation results, which validates the generalization capability of the proposed architecture to never-seen environment locations without site-specific overfitting for planned-real-world deployment across various agricultural backgrounds with differing

Table 12. Leave-Site-Out 5-Fold Cross-Validation

Fold	Accuracy	Latency ms	Model size mb	Training time min	Train sample	Val samples
1	0.9473	14.7271	85.53	36.29	7281	1896
2	0.9482	14.0806	85.53	31.78	7322	1855
3	0.9514	16.2871	85.53	33.65	7325	1852
4	0.9373	13.3393	85.53	34.24	7325	1852
5	0.9489	15.7702	85.53	34.33	7455	1722
Metric	Mean	Std Dev	Min	Max	95% CI	CI Range
Accuracy	0.9466	0.0054	0.9374	0.9514	± 0.0047	[0.9419, 0.9514]
Latency Ms	14.84	1.20	13.34	16.29	± 1.06	[13.79, 15.90]
Model Size Mb	85.53	0.00	85.53	85.53	-	-

environmental conditions, camera hardware setups, and fruit characteristics.

Both confusion patterns and confidence distributions provide insight into how the model behaves and where it is likely to fail. Example predictions. For each example dataset, a representative sample of prediction classes is shown in Fig. 8, depicting successful classifications and scenario failure cases.

To enhance reproducibility, we provide the following simplified pseudo-code based on the Integration of Hybrid GA-PSO-SA with Self-Adaptive Dynamic Attention for Multi-Objective CNN-Based architecture, as presented in Table 13.

In the Table 14. The ablation study, conducted

with seed 40, reveals an unusual path in which architectural complexity doesn't correlate with better performance. Starting with the CNN baseline, which achieves 70.87% accuracy, we only see small improvements with the addition of CBAM. This yields an accuracy of 71.53% and an F1 score of 72.74%, suggesting that attention mechanisms are of little use for classifying palm oil fruit, given the fruit cluster's homogeneous visual features. The integration of a genetic algorithm is the first significant step forward in performance. It increases accuracy to 77.50%, a 6.63% improvement, underscoring the importance of hyperparameter tuning over additional architectural complexity. The best configuration, CNN+GA+PSO, achieved 95.73% accuracy and 95.84% F1 score. This highlights the power of different optimization approaches working together, with GA's comprehensive search and PSO's local search, to accurately and efficiently navigate the hyperparameter space to derive the ideal bias-variance trade-off, as shown by the small 0.16% precision-recall gap, which means the model does not overfit.

Certainly, adding more elements in subsequent steps will lead to noticeable degradation in achieving optimal performance goals in metaheuristic optimization. In simulated annealing, the metaheuristic reduces the depth of accuracy by 95.07%, indicating that the convergence trajectories from GA and PSO move towards lower accuracy. When CBAM is layered on top of CNN+GA+PSO+SA, the decline in overall accuracy exceeds 94.58% at low complexity (1.15% below peak performance). In 9 or more dimensions, the space traps the optimizer in local optima. In the end, the configuration of the final



Figure. 8 Example of Correctly Classified Oil Palm Fruit Images

Table 13. Pseudo-Code Integrated Architecture Pipeline
Algorithm

Integration of Hybrid GA-PSO-SA with Self-Adaptive Dynamic Attention for Multi-Objective CNN-Based
Input: Population size N_{pop} , generations N_{gen} , archive size $N_{archive}$ Output: Pareto front A 1: Initialize population $P = \{x_1, \dots, x_{N_{pop}}\}$ uniformly in $[0,1]^{16}$ 2: Evaluate $F(x_i)$ for all $x_i \in P$ 3: Perform fast non-dominated sort to get fronts $F_0, F_1, \dots$ 4: Calculate crowding distance for each front 5: Initialize Pareto archive $A \leftarrow F_0[1:N_{archive}]$ 6: Set temperature $T \leftarrow 0.1$ 7: for $t = 1$ to N_{gen} do 8: // PSO Phase 9: for each particle $x_i \in P$ do 10: Select g_leader randomly from A 11: Update velocity: $v_i \leftarrow w \cdot v_i + c_1 \cdot r_1 \odot (p_i - x_i) + c_2 \cdot r_2 \odot (g_leader - x_i)$ 12: Update position: $x_i \leftarrow \text{clip}(x_i + v_i, 0, 1)$ 13: end for 14: // GA Phase 15: Perform fast non-dominated sort on P 16: Select elites E (top 50% by rank and crowding distance) 17: Generate offspring O through uniform crossover and mutation 18: Combine: $P_{new} \leftarrow E \cup O$ 19: // Evaluation 20: Evaluate $F(x)$ for all $x \in P_{new}$ 21: // SA Phase 22: $P_{accepted} \leftarrow \{\}$ 23: for each $x_{new} \in P_{new}$ do 24: if $ P_{accepted} < N_{pop}$ then 25: $P_{accepted} \leftarrow P_{accepted} \cup \{x_{new}\}$ 26: else 27: $x_{worst} \leftarrow \text{argmin}_{\{x \in P_{accepted}\}} f_1(x)$ 28: $\Delta \leftarrow \text{composite_score}(x_{new}, x_{worst})$ 29: if $\Delta > 0$ or $\text{rand}() < \exp(\Delta/T)$ then 30: $P_{accepted} \leftarrow (P_{accepted} \setminus \{x_{worst}\}) \cup \{x_{new}\}$ 31: end if 32: end if 33: end for 34: $P \leftarrow P_{accepted}$ 35: // Update Pareto Archive 36: Perform fast non-dominated sort on $P \cup A$ 37: Calculate crowding distance for front F_0 38: $A \leftarrow F_0[1:N_{archive}]$ sorted by crowding distance 39: $HV \leftarrow \text{calculate_hypervolume}(A)$ 40: // Cool temperature 41: $T \leftarrow 0.9 \cdot T$ 42: end for 43: return A

Table 14. Ablation study results on validation data measured using Model Performance Metrics

Model	Accuracy %	Precision %	Recall %	F1-Score %
CNN Baseline	70.87	73.24	73.31	70.87
CNN+CBAM	71.53	74.12	74.46	72.74
CNN+GA	77.50	78.16	78.05	77.50
CNN+GA+PSO	95.73	95.94	95.78	95.84
CNN+GA+PSO+SA	95.07	95.41	95.17	95.22
CNN+GA+PSO+SA+CBAM	94.58	94.75	94.73	94.71
CNN+GA+PSO+SA+SADA	96.86	96.03	95.94	95.97

CNN+GA+PSO+SA+SADA, which attained an outright accuracy of 96.86%, yields a 95.97% F1 score, which increases the CNN+GA+PSO to 95.84%; hence, the imbalance in optimization indicates overfitting. Occam's razor in deep learning focuses on evidence rather than the complexity of the deep architecture; hence, optimal performance results from simpler structures and more effective dual meta-heuristic optimization. To validate these findings and ensure statistical robustness, the optimal CNN+GA+PSO configuration was rerun 10 times with 10 different random seeds, achieving a mean confidence interval test accuracy of 96.12% as documented in Table 21, which confirms the model's stability and generalization capability across varied initialization conditions.

In addition, we include a comparison of the baseline models which can be seen in Table 15.

In Table 15. For all five instances of the evaluation of model performance on all five seeds (42, 123, 456, 789, and 1024). The overall detection accuracy of the GA-PSO-SA+SADA was 96.12%, the highest achieved, with a latency of 2.61 ± 0.01 ms, improving ViT-tiny by a factor of 4.1 in terms of speed, and MobileNetV2 and ResNet50 by factors of 4.4 and 5.1, respectively. That was better than any of the baseline models. The proposed model, which is 43.64 MB in size, also achieved higher accuracy than both MobileNetV2 (95.53%) and ResNet50 (95.27%), which are considered good models. It also achieved higher accuracy than the ViT-tiny model, which reached 96.06% with a much narrower confidence interval. The hybrid optimisation framework achieved the Pareto optimal for the models' accuracy, latency, and size. It also improved latency by 4.1 times (more than 4.1 times faster than ViT-tiny) while maintaining the same or better accuracy, making it a good choice for edge devices used in

Table 15. Baseline Comparison Table Confidence Intervals

Model	Run	Seed	Mean CI Test Accuracy	Accuracy Range CI	Latency (ms)	Model Size (MB)
MobileNetV2	10	[42, 123, 456, 789, 1024,2048,3072,4096, 5120, 6144]	95.53	[95.22, 95.85]	11.51 ± 0.62	8.5
ResNet50	10	[42, 123, 456, 789, 1024,2048,3072,4096, 5120, 6144]	95.27	[95.03, 95.51]	13,37 ± 0,27	89.7
ViT-tiny	10	[42, 123, 456, 789, 1024,2048,3072,4096, 5120, 6144]	96.06	[95.53, 96.59]	10.74 ± 0.07	21.08
Proposed (GA-PSO-SA+SADA)	10	[42, 123, 456, 789, 1024,2048,3072,4096, 5120, 6144]	96.12	[95.85, 96.39]	2.61 ± 0,01	43.64

Precision Agriculture to classify ripeness in real time.

Table 16. The SOTA comparison chart shows a significant change in object detection between 2024 and 2025. HAT-YOLOV8 leads with a top score of 88.9 but with a cost of 220 ms/fps. Transformer models, hybrid ones like EfficientNet B3 and MobileNetV3 Large, get 91 92 but with high latency 152 491 ms and large size 45 147 MB. The YOLO series v8s v10n consistently scored 76 81 with a good latency of 15 280 ms, proving the YOLO design is still useful for real-time work. PBR-YOLO (Jetson Orin NX) is the best choice for edge computing, with 77.6% accuracy, super-low latency of 15.8 ms, and a small size of 2.7 MB, beating all rivals on efficiency.

EEGNet (2025) is a strong choice with 76.23% accuracy, 15.7 ms latency, and 6.3 MB size, making it well-suited for low-resource settings. The trend is

clear: heavy models like EfficientNet and MobileNet aim for high accuracy with no limits on what can be used, while YOLO and PBR-YOLO versions aim to balance high performance with efficiency for real-world use in gadgets and self-driving cars.

The performance improvements from Self-Adaptive Dynamic Attention in the hybrid GA-PSO-SA mechanism confirm the configuration's value, and PSO + GA surpasses a simple CNN with CBAM. Post hoc analysis with seed 40 focused on CNN architecture and hyper-parameter tuning, disengaged from GA without the predicted outcome. Performance remained below plausible with CNN + CBAM, achieving only 71.53% accuracy, with a 0.81% to 0.88% range around the balanced accuracy

Table 16. Comparison SOTA

Method	Year	Accuracy	Latency (ms/fps)	Size (MB)
EEGNet [26]	2025	76.23	15.7	6.3
Faster R-CNN [27]	2024	78.7	18	113.7
YOLOv8s [27]	2024	81.3	280	26.5
ViT [27]	2024	81.1	120	54.8
HCFormer [27]	2024	88.9	220	35.7
Faster R-CNN [28]	2025	75.5	35.6	113.7
YOLOv10n [28]	2025	76.9	8.2	5.8
PBR-YOLO (Jetson Orin NX) [28]	2025	77.6	15.8	2.7
Faster R-CNN (ResNet-101) [29]	2025	78.7	18	85.2
YOLOv8s (CSPNet) [29]	2025	81.3	280	26.5
ViT (ViT-B 16) [29]	2025	81.1	120	54.8
HAT-YOLOV8 [29]	2025	88.9%	220	35.7
EfficientNet-B3 [30]	2025	92.64	491.58	147.34
MobileNetV3 Large [30]	2025	91.48	152.24	45.62
TL-HACNN+EfficientNet-B3 [30]	2025	91.25	280.67	59.81
TL-HACNN+MobileNetV3 Large [30]	2025	91.04	86.94	26.52

mean. Transformed hyper-parameter optimization with CNN + GA + clustered PSO at 95.73% accuracy and 95.84% F1 score shows 18.23% difference from baseline, confirming the Coef of traffic. These theorists demonstrated that metaheuristic rules of architecture were less well theorized, which drove GA/CB PSO. Novelty here: the other half of GA isolation. Adaptations sustained the dual search driving hyperparameters to an optimal, balanced system without overfitting.

Adding more components (SA, CBAM, SADA) systematically deteriorates performance, revealing the curse of dimensionality in metaheuristic optimization. When Simulated Annealing (SA) was added, the accuracy decreased to 95.07%, indicating that the acceptance mechanism in SA probabilistically disrupts, rather than aids, the convergence of the GA-PSO trajectory. Further performance drops were evident when CBAM was added to the architecture, combined with CNN+GA+PSO+SA, resulting in an accuracy of 94.58%, a 1.15% drop from the highest performance. This is due to the exploding search space introduced by adding 9 to 25+ dimensions, causing the optimizer to get stuck in poor local optima. The final architecture of CNN+GA+PSO+SA+SADA was able to reach the highest raw accuracy of 96.86%, but the decline of the F1-score to 95.97% (versus 95.84% of CNN+GA+PSO) indicates optimization was unstable, a sign of the system being overfitted to the training data, where the improved accuracy came at the cost of poor generalization of the model. The results presented clearly demonstrate the phenomenon of Occam's Razor in deep learning. Simpler architectures with more effective dual metaheuristic optimization will clearly outperform complex architectures with inferior optimization.

The total mean test accuracy of $96.12\% \pm 0.39\%$ and a 95% confidence interval of $[95.85\%, 96.39\%]$ indicate the model's stability and generalization capability, which is validated through robustness testing across 10 independent runs with different random seeds. If only a mean performance perspective is regarded, then the test accuracy of 96.12% is 0.94% greater than the validation accuracy of 95.18%, and this pattern is noticeably unchanged in the best performing run (run 1), where the test accuracy of 96.86% is greater than the validation accuracy of 96.10% by 0.76%, and is consistent here as well. These gaps are uncharacteristic of the generalization gaps 3-5% reported in literature, suggesting a high (based on literature) capability in generalization has been achieved. The mean accuracy, validated through Leave-Site-Out 5-fold cross-validation, is $94.66\% \pm 0.54\%$ across the inaccessible

underlying geographical sites, thereby corroborating that the learned ripeness features are not geographically localized but generalizable, further validating spatial generalization. The pareto-optimal analysis bestows practical deployment flexibility, where, with the BALANCED criterion, Pareto_2 (ResNet18: 88.61% accuracy, 4.93ms latency, 43.6MB) is chosen as the optimum trade-off, outperforming all baseline models (MobileNetV2: 95.53%, ResNet50: 95.27%) in both accuracy and computational efficiency, while achieving $4.1\times$ faster inference than ViT-tiny, making it most appropriate for deployment in real-time edge-devices in oil palm plantations.

5. Conclusion

In developing a framework that integrates a hybrid GA-PSO SA metaheuristic optimization with self-adaptive dynamic attention mechanisms within a multi-tier optimization, this work has begun to bridge the gap in the automatic classification of oil-palm fruit ripeness stages by proposing methods that have not been previously documented. A relative CNN+GA+PSO balance was achieved at 95.73 percent for accuracy, 95.84 percent for F1-score, and 0.16 percent for the precision-recall gap, validated across 10 distinct runs with a mean test accuracy of $96.12\% \pm 0.39\%$. Furthermore, the ablation study demonstrated that complementing dual search methods hyperparameter optimization was far superior in impact than simply engineering the architecture of the network, where GA-PSO produced 18.23 percent more improvement than the single algorithms within the pseudosingle approach and was more stable in convergence relative to the other complex topologies suffering the curse of dimensionality.

The multi-objective Pareto optimization framework has successfully balanced trade-offs among accuracy, inference latency, and model size, yielding deployment-ready solutions despite highly variable operational constraints. Pareto_2, the best-balanced model, achieves an accuracy of 88.61%, a latency of 4.93ms, and a size of 43.6 MB, resulting in $4.1\times$ faster inference than ViT-tiny and surpassing the accuracy of MobileNetV2 (95.53%) and ResNet50 (95.27%). The Leave-Site-Out 5-fold cross-validation attains an uncommonly high accuracy of $94.66\% \pm 0.54\%$ with unseen geographical sites, in combination with test-validation gaps (0.76-0.94%) substantially lower than typical literature gaps (3-5%), confirming remarkable spatial generalization, and is robust to a wide range of environmental changes and intra-class variance, camera hardware differences,

and the varied fruit characteristics found in Indonesian plantations.

The oil palm sector finds significant value in implementing the advancements from this research. Compared with previous manual inspection methods, which achieved 65-75% accuracy and 15-20% harvest loss, translating into losses of billions of rupiah annually, the proposed framework offers real-time ripeness detection with 96.86% peak accuracy and 2.61ms latency. The compacted model (43.564 MB) with improved inference characteristics will provide a Pareto-optimal set, enabling flexible model selection based on the system's high-efficiency or high-accuracy requirements. Future research should be geared towards: (1) the exploration of multi-modal fusion methods that combine RGB imaging with near- infrared spectroscopy and thermal imaging to go beyond visual characteristics of maturity to include biochemical features, (2) the establishment of federated learning systems that address non-iid data distribution and data privacy to allow for the seamless updating of models across various plantational environments, (3) the use of explainable AI to aid decision making in Ag site practices in the form of Grad- CAM or SHAP, (4) the timely structural architectural modeling of ripening and the integration of video content, and (5) through the tracking of the economic impact on the key oil content of harvested crop, the validation of automated harvest mobile applications robotics with field use.

These supplementary modules will also enhance and make comprehensive the full smart farming ecosystems already active within the palm oil sector, and maximize value chain optimization from farming to processing.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

Usman collected, designed, and gathered research data, analyzed data, experiments, and tests, and contributed to the writing of the manuscript. Fatchul Arifin and Rustam Asnawi acted as research supervisors and mentors. Usman validated the test data and analyzed the data. All authors have approved the final version.

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